3AL Hand-in 3 2024 Solutions

Due: Friday 19 April 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.

   (a) If G is a finite group, then $|\text{class}\, a|$ divides $|G|$ for all $a \in G$.

   (b) If $H \lhd G$ with $H \neq G$, then $H = N_G(H)$.

   (c) If G is a group, then $\langle g \rangle \subseteq C_G(g)$ for all $g \in G$.

   (d) If G is a group and $a, b \in G$, then ab and ba are conjugate.

(a) True. $|\text{class}\, a| = |G : C_G(a)|$, which must divide $|G|$ by Lagrange's Theorem.

(b) False. If $H \lhd G$, then $\mathbb{N}_G(H) = G \neq H$.

(c) True. Let $g^n \in \langle g \rangle$, then $g^n g = g^{n+1} = gg^{n+1}$, thus $g^n \in C_G(g)$.

(d) True. Notice that $a^{-1} \in G$, so we can conjugate ab by a^{-1} to get ba.

[4 marks]

2. Let G be a group and $g \in G$. Suppose $|g| = n$. Prove that if g is the only element of G with order n, then $g \in Z(G)$.

Think about conjugacy classes: If g is the only element of G with order n, then $\text{class}\, g = \{g\}$. Thus g has a singleton conjugacy class and so $g \in Z(G)$.

[4 marks]

3. Let G be a group and T a set of representatives of the distinct conjugacy classes of G. Prove that $G = \bigcup_{g \in G} gTg^{-1}$.

Since G is a group, we must have that the given union is contained in G. Let $a \in G$, we need to show that there is some $g \in G$ such that $a \in gTg^{-1}$. Since $a \in G$, there is some $b \in T$ such that $a \in \operatorname{class} b$. That is, there is some $c \in G$ such that $a = cbc^{-1}$. Taking $g = c$, we are done.

[4 marks]

Maximum mark = 10

Available marks = 12